



MATH140: Differentiation Rules

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Chain rule

1. A cost function is $C(x) = (1x + 1)^3$. Find $C'(1)$ using the chain rule.

$$f(x) = (1x + 1)^3, x = 1$$

Answer: _____

2. A cost function is $C(x) = (1x + 7)^2$. Find $C'(1)$ using the chain rule.

$$f(x) = (1x + 7)^2, x = 1$$

Answer: _____

3. A cost function is $C(x) = (4x + 5)^3$. Find $C'(1)$ using the chain rule.

$$f(x) = (4x + 5)^3, x = 1$$

Answer: _____

4. A cost function is $C(x) = (2x + 8)^3$. Find $C'(3)$ using the chain rule.

$$f(x) = (2x + 8)^3, x = 3$$

Answer: _____

5. A cost function is $C(x) = (2x + 4)^2$. Find $C'(0)$ using the chain rule.

$$f(x) = (2x + 4)^2, x = 0$$

Answer: _____

6. A cost function is $C(x) = (2x + 2)^2$. Find $C'(3)$ using the chain rule.

$$f(x) = (2x + 2)^2, x = 3$$

Answer: _____



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Labanan ang agwat.

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7. A cost function is $C(x) = (1x + 5)^3$. Find $C'(1)$ using the chain rule.

$$f(x) = (1x + 5)^3, x = 1$$

Answer: _____

8. A cost function is $C(x) = (1x + 5)^3$. Find $C'(3)$ using the chain rule.

$$f(x) = (1x + 5)^3, x = 3$$

Answer: _____

Differentiating polynomials

9. Total revenue is $R(x) = 1x^3 + -2x^2 + 14x + 59$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 1x^3 + -2x^2 + 14x + 59, x = 2$$

Answer: _____

10. Total revenue is $R(x) = 2x^3 + -4x^2 + 23x + 70$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 2x^3 + -4x^2 + 23x + 70, x = 2$$

Answer: _____

11. Total revenue is $R(x) = 1x^3 + 0x^2 + 27x + 20$. Find the marginal revenue $MR = R'(1)$.

$$f(x) = 1x^3 + 0x^2 + 27x + 20, x = 1$$

Answer: _____

12. Total revenue is $R(x) = 1x^3 + -3x^2 + 14x + 94$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 1x^3 + -3x^2 + 14x + 94, x = 2$$

Answer: _____

13. Total revenue is $R(x) = 1x^3 + -4x^2 + 30x + 58$. Find the marginal revenue $MR = R'(1)$.

$$f(x) = 1x^3 + -4x^2 + 30x + 58, x = 1$$

Answer: _____

14. Total revenue is $R(x) = 2x^3 + -1x^2 + 19x + 32$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 2x^3 + -1x^2 + 19x + 32, x = 2$$

Answer: _____



15. Total revenue is $R(x) = 1x^3 + -4x^2 + 23x + 59$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 1x^3 + -4x^2 + 23x + 59, x = 2$$

Answer: _____

16. Total revenue is $R(x) = 2x^3 + 0x^2 + 29x + 1$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 2x^3 + 0x^2 + 29x + 1, x = 2$$

Answer: _____

Power rule

17. Find the derivative of $f(x) = 7x^2$ and evaluate $f'(2)$.

$$f(x) = 7x^2, x = 2$$

Answer: _____

18. Find the derivative of $f(x) = 6x^3$ and evaluate $f'(3)$.

$$f(x) = 6x^3, x = 3$$

Answer: _____

19. Find the derivative of $f(x) = 5x^2$ and evaluate $f'(4)$.

$$f(x) = 5x^2, x = 4$$

Answer: _____

20. Find the derivative of $f(x) = 1x^5$ and evaluate $f'(3)$.

$$f(x) = 1x^5, x = 3$$

Answer: _____

21. Find the derivative of $f(x) = 5x^4$ and evaluate $f'(2)$.

$$f(x) = 5x^4, x = 2$$

Answer: _____

22. Find the derivative of $f(x) = 8x^2$ and evaluate $f'(4)$.

$$f(x) = 8x^2, x = 4$$

Answer: _____



23. Find the derivative of $f(x) = 8x^2$ and evaluate $f'(2)$.

$$f(x) = 8x^2, \quad x = 2$$

Answer: _____

Product rule

24. Revenue is $R(x) = (3x + 7)(4x + 5)$. Use the product rule to find $R'(0)$.

$$f(x) = (3x + 7)(4x + 5), \quad x = 0$$

Answer: _____

25. Revenue is $R(x) = (1x + 6)(4x + 2)$. Use the product rule to find $R'(2)$.

$$f(x) = (1x + 6)(4x + 2), \quad x = 2$$

Answer: _____

26. Revenue is $R(x) = (5x + 4)(4x + 9)$. Use the product rule to find $R'(3)$.

$$f(x) = (5x + 4)(4x + 9), \quad x = 3$$

Answer: _____

27. Revenue is $R(x) = (1x + 6)(5x + 10)$. Use the product rule to find $R'(0)$.

$$f(x) = (1x + 6)(5x + 10), \quad x = 0$$

Answer: _____

28. Revenue is $R(x) = (4x + 3)(5x + 4)$. Use the product rule to find $R'(2)$.

$$f(x) = (4x + 3)(5x + 4), \quad x = 2$$

Answer: _____

29. Revenue is $R(x) = (4x + 5)(3x + 9)$. Use the product rule to find $R'(1)$.

$$f(x) = (4x + 5)(3x + 9), \quad x = 1$$

Answer: _____

30. Revenue is $R(x) = (1x + 2)(2x + 2)$. Use the product rule to find $R'(4)$.

$$f(x) = (1x + 2)(2x + 2), \quad x = 4$$

Answer: _____





MATH140: Differentiation Rules

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ANSWER KEY & SOLUTIONS

Topics: Power rule, Product rule, Chain rule, Differentiating polynomials. All answers verified by independent computation.

Solutions



Chain rule

1. A cost function is $C(x) = (1x + 1)^3$. Find $C'(1)$ using the chain rule.

$$f(x) = (1x + 1)^3, \quad x = 1$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 3 \cdot 1(1x + 1)^2, \quad f'(1) = 12$

2. A cost function is $C(x) = (1x + 7)^2$. Find $C'(1)$ using the chain rule.

$$f(x) = (1x + 7)^2, \quad x = 1$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 2 \cdot 1(1x + 7)^1, \quad f'(1) = 16$

3. A cost function is $C(x) = (4x + 5)^3$. Find $C'(1)$ using the chain rule.

$$f(x) = (4x + 5)^3, \quad x = 1$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 3 \cdot 4(4x + 5)^2, \quad f'(1) = 972$

4. A cost function is $C(x) = (2x + 8)^3$. Find $C'(3)$ using the chain rule.

$$f(x) = (2x + 8)^3, \quad x = 3$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 3 \cdot 2(2x + 8)^2, \quad f'(3) = 1176$

5. A cost function is $C(x) = (2x + 4)^2$. Find $C'(0)$ using the chain rule.

$$f(x) = (2x + 4)^2, \quad x = 0$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 2 \cdot 2(2x + 4)^1, \quad f'(0) = 16$



6. A cost function is $C(x) = (2x + 2)^2$. Find $C'(3)$ using the chain rule.

$$f(x) = (2x + 2)^2, \quad x = 3$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 2 \cdot 2(2x + 2)^1, \quad f'(3) = 32$

7. A cost function is $C(x) = (1x + 5)^3$. Find $C'(1)$ using the chain rule.

$$f(x) = (1x + 5)^3, \quad x = 1$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 3 \cdot 1(1x + 5)^2, \quad f'(1) = 108$

8. A cost function is $C(x) = (1x + 5)^3$. Find $C'(3)$ using the chain rule.

$$f(x) = (1x + 5)^3, \quad x = 3$$

→ Chain rule: $d/dx[(ax+b)^n] = n \cdot a \cdot (ax+b)^{(n-1)}$.

→ $C'(x) = \{n\} \cdot \{a\} \cdot (\{a\}x + \{b\})^{\{new_n\}}$.

→ $C'(\{x_val\}) = \{n\} \cdot \{a\} \cdot (\{inner\})^{\{new_n\}} = \{answer_deriv_chain210\}$.

Answer: $f'(x) = 3 \cdot 1(1x + 5)^2, \quad f'(3) = 192$



Differentiating polynomials

9. Total revenue is $R(x) = 1x^3 + -2x^2 + 14x + 59$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 1x^3 + -2x^2 + 14x + 59, x = 2$$

$$\rightarrow MR = R'(x) = 3x^2 + -4x + 14.$$

$$\rightarrow R'(2) = 3(2)^2 + -4(2) + 14 = 18.$$

Answer: $f'(x) = 3x^2 + -4x + 14, f'(2) = 18$

10. Total revenue is $R(x) = 2x^3 + -4x^2 + 23x + 70$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 2x^3 + -4x^2 + 23x + 70, x = 2$$

$$\rightarrow MR = R'(x) = 6x^2 + -8x + 23.$$

$$\rightarrow R'(2) = 6(2)^2 + -8(2) + 23 = 31.$$

Answer: $f'(x) = 6x^2 + -8x + 23, f'(2) = 31$

11. Total revenue is $R(x) = 1x^3 + 0x^2 + 27x + 20$. Find the marginal revenue $MR = R'(1)$.

$$f(x) = 1x^3 + 0x^2 + 27x + 20, x = 1$$

$$\rightarrow MR = R'(x) = 3x^2 + 0x + 27.$$

$$\rightarrow R'(1) = 3(1)^2 + 0(1) + 27 = 30.$$

Answer: $f'(x) = 3x^2 + 0x + 27, f'(1) = 30$

12. Total revenue is $R(x) = 1x^3 + -3x^2 + 14x + 94$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 1x^3 + -3x^2 + 14x + 94, x = 2$$

$$\rightarrow MR = R'(x) = 3x^2 + -6x + 14.$$

$$\rightarrow R'(2) = 3(2)^2 + -6(2) + 14 = 14.$$

Answer: $f'(x) = 3x^2 + -6x + 14, f'(2) = 14$

13. Total revenue is $R(x) = 1x^3 + -4x^2 + 30x + 58$. Find the marginal revenue $MR = R'(1)$.

$$f(x) = 1x^3 + -4x^2 + 30x + 58, x = 1$$

$$\rightarrow MR = R'(x) = 3x^2 + -8x + 30.$$

$$\rightarrow R'(1) = 3(1)^2 + -8(1) + 30 = 25.$$

Answer: $f'(x) = 3x^2 + -8x + 30, f'(1) = 25$

14. Total revenue is $R(x) = 2x^3 + -1x^2 + 19x + 32$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 2x^3 + -1x^2 + 19x + 32, x = 2$$

$$\rightarrow MR = R'(x) = 6x^2 + -2x + 19.$$

$$\rightarrow R'(2) = 6(2)^2 + -2(2) + 19 = 39.$$

Answer: $f'(x) = 6x^2 + -2x + 19, f'(2) = 39$



15. Total revenue is $R(x) = 1x^3 + -4x^2 + 23x + 59$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 1x^3 + -4x^2 + 23x + 59, \quad x = 2$$

$$\rightarrow MR = R'(x) = 3x^2 + -8x + 23.$$

$$\rightarrow R'(2) = 3(2)^2 + -8(2) + 23 = 19.$$

Answer: $f'(x) = 3x^2 + -8x + 23, \quad f'(2) = 19$

16. Total revenue is $R(x) = 2x^3 + 0x^2 + 29x + 1$. Find the marginal revenue $MR = R'(2)$.

$$f(x) = 2x^3 + 0x^2 + 29x + 1, \quad x = 2$$

$$\rightarrow MR = R'(x) = 6x^2 + 0x + 29.$$

$$\rightarrow R'(2) = 6(2)^2 + 0(2) + 29 = 53.$$

Answer: $f'(x) = 6x^2 + 0x + 29, \quad f'(2) = 53$



Power rule

17. Find the derivative of $f(x) = 7x^2$ and evaluate $f'(2)$.

$$f(x) = 7x^2, \quad x = 2$$

$$\rightarrow f'(x) = 14x^1 \text{ (power rule).}$$

$$\rightarrow f'(2) = 14(2)^1 = 28.$$

Answer: $f'(x) = 14x^1, \quad f'(2) = 28$

18. Find the derivative of $f(x) = 6x^3$ and evaluate $f'(3)$.

$$f(x) = 6x^3, \quad x = 3$$

$$\rightarrow f'(x) = 18x^2 \text{ (power rule).}$$

$$\rightarrow f'(3) = 18(3)^2 = 162.$$

Answer: $f'(x) = 18x^2, \quad f'(3) = 162$

19. Find the derivative of $f(x) = 5x^2$ and evaluate $f'(4)$.

$$f(x) = 5x^2, \quad x = 4$$

$$\rightarrow f'(x) = 10x^1 \text{ (power rule).}$$

$$\rightarrow f'(4) = 10(4)^1 = 40.$$

Answer: $f'(x) = 10x^1, \quad f'(4) = 40$

20. Find the derivative of $f(x) = 1x^5$ and evaluate $f'(3)$.

$$f(x) = 1x^5, \quad x = 3$$

$$\rightarrow f'(x) = 5x^4 \text{ (power rule).}$$

$$\rightarrow f'(3) = 5(3)^4 = 405.$$

Answer: $f'(x) = 5x^4, \quad f'(3) = 405$

21. Find the derivative of $f(x) = 5x^4$ and evaluate $f'(2)$.

$$f(x) = 5x^4, \quad x = 2$$

$$\rightarrow f'(x) = 20x^3 \text{ (power rule).}$$

$$\rightarrow f'(2) = 20(2)^3 = 160.$$

Answer: $f'(x) = 20x^3, \quad f'(2) = 160$

22. Find the derivative of $f(x) = 8x^2$ and evaluate $f'(4)$.

$$f(x) = 8x^2, \quad x = 4$$

$$\rightarrow f'(x) = 16x^1 \text{ (power rule).}$$

$$\rightarrow f'(4) = 16(4)^1 = 64.$$

Answer: $f'(x) = 16x^1, \quad f'(4) = 64$



23. Find the derivative of $f(x) = 8x^2$ and evaluate $f'(2)$.

$$f(x) = 8x^2, \quad x = 2$$

$$\rightarrow f'(x) = 16x^1 \text{ (power rule).}$$

$$\rightarrow f'(2) = 16(2)^1 = 32.$$

Answer: $f'(x) = 16x^1, \quad f'(2) = 32$



Product rule

24. Revenue is $R(x) = (3x + 7)(4x + 5)$. Use the product rule to find $R'(0)$.

$$f(x) = (3x + 7)(4x + 5), \quad x = 0$$

$$\rightarrow u = 3x+7, \quad v = 4x+5. \quad u' = 3, \quad v' = 4.$$

$$\rightarrow R'(x) = u'v + uv' = 3(4x+5) + 4(3x+7).$$

$$\rightarrow R'(0) = 3(5) + 4(7) = 43.$$

Answer: $f'(0) = 3 \cdot 5 + 4 \cdot 7 = 43$

25. Revenue is $R(x) = (1x + 6)(4x + 2)$. Use the product rule to find $R'(2)$.

$$f(x) = (1x + 6)(4x + 2), \quad x = 2$$

$$\rightarrow u = 1x+6, \quad v = 4x+2. \quad u' = 1, \quad v' = 4.$$

$$\rightarrow R'(x) = u'v + uv' = 1(4x+2) + 4(1x+6).$$

$$\rightarrow R'(2) = 1(10) + 4(8) = 42.$$

Answer: $f'(2) = 1 \cdot 10 + 4 \cdot 8 = 42$

26. Revenue is $R(x) = (5x + 4)(4x + 9)$. Use the product rule to find $R'(3)$.

$$f(x) = (5x + 4)(4x + 9), \quad x = 3$$

$$\rightarrow u = 5x+4, \quad v = 4x+9. \quad u' = 5, \quad v' = 4.$$

$$\rightarrow R'(x) = u'v + uv' = 5(4x+9) + 4(5x+4).$$

$$\rightarrow R'(3) = 5(21) + 4(19) = 181.$$

Answer: $f'(3) = 5 \cdot 21 + 4 \cdot 19 = 181$

27. Revenue is $R(x) = (1x + 6)(5x + 10)$. Use the product rule to find $R'(0)$.

$$f(x) = (1x + 6)(5x + 10), \quad x = 0$$

$$\rightarrow u = 1x+6, \quad v = 5x+10. \quad u' = 1, \quad v' = 5.$$

$$\rightarrow R'(x) = u'v + uv' = 1(5x+10) + 5(1x+6).$$

$$\rightarrow R'(0) = 1(10) + 5(6) = 40.$$

Answer: $f'(0) = 1 \cdot 10 + 5 \cdot 6 = 40$

28. Revenue is $R(x) = (4x + 3)(5x + 4)$. Use the product rule to find $R'(2)$.

$$f(x) = (4x + 3)(5x + 4), \quad x = 2$$

$$\rightarrow u = 4x+3, \quad v = 5x+4. \quad u' = 4, \quad v' = 5.$$

$$\rightarrow R'(x) = u'v + uv' = 4(5x+4) + 5(4x+3).$$

$$\rightarrow R'(2) = 4(14) + 5(11) = 111.$$

Answer: $f'(2) = 4 \cdot 14 + 5 \cdot 11 = 111$

29. Revenue is $R(x) = (4x + 5)(3x + 9)$. Use the product rule to find $R'(1)$.

$$f(x) = (4x + 5)(3x + 9), \quad x = 1$$

$$\rightarrow u = 4x+5, \quad v = 3x+9. \quad u' = 4, \quad v' = 3.$$

$$\rightarrow R'(x) = u'v + uv' = 4(3x+9) + 3(4x+5).$$

$$\rightarrow R'(1) = 4(12) + 3(9) = 75.$$

Answer: $f'(1) = 4 \cdot 12 + 3 \cdot 9 = 75$



30. Revenue is $R(x) = (1x + 2)(2x + 2)$. Use the product rule to find $R'(4)$.

$$f(x) = (1x + 2)(2x + 2), \quad x = 4$$

$$\rightarrow u = 1x+2, v = 2x+2. \quad u' = 1, v' = 2.$$

$$\rightarrow R'(x) = u'v + uv' = 1(2x+2) + 2(1x+2).$$

$$\rightarrow R'(4) = 1(10) + 2(6) = 22.$$

Answer: $f'(4) = 1 \cdot 10 + 2 \cdot 6 = 22$

